

Commutator

Def. Let G be a group. Given $x, y \in G$, define

$$[x, y] \stackrel{\text{def}}{=} x^{-1}y^{-1}xy$$

is called the commutator of x and y . And, given two subset A and B ,

$$[A, B] \stackrel{\text{def}}{=} \langle [a, b] \mid a \in A, b \in B \rangle$$

Particularly, $G' \stackrel{\text{def}}{=} [G, G]$ is called the derived set.

Prop. G' char G , (so $G' \trianglelefteq G$).

Proof. Given $\sigma \in \text{Aut}(G)$ and $[x, y]$ for arbitrary $x, y \in G$,

$$\sigma([x, y]) = \sigma(x^{-1}y^{-1}xy) = \sigma(x)^{-1} \cdot \sigma(y)^{-1} \cdot \sigma(x) \cdot \sigma(y) = [\sigma(x), \sigma(y)]$$

$$\text{So, } \sigma[[G, G]] = [G, G].$$

Prop. Given $H \leq G$, $H \trianglelefteq G$ if and only if $[H, G] \leq H$

Proof. $H \trianglelefteq G \Leftrightarrow$ for all $g \in G$ and $h \in H$, $ghg^{-1} \in H$

$$[H, G] \leq H \Leftrightarrow \text{for all } g \in G \text{ and } h \in H, [h, g] = h^{-1}g^{-1}hg \in H.$$

So proved.

Prop. Given $N \trianglelefteq G$, G/N is abelian if and only if $G' \leq N$.

Proof. G/N is abelian iff for any $xN, yN \in G/N$, $xyN = yxN$

iff for any $x, y \in G$, $x^{-1}y^{-1}xy \in N$

iff $G' \leq N$.

Note: G' is the smallest normal subgroup of G s.t. the factor is abelian.

Note: If G is abelian, then $G' = 1$.

$$\text{Note: } [x, y] \cdot [y, z] = x^{-1}y^{-1}xy \cdot y^{-1}z^{-1}yz = x^{-1}y^{-1}xz^{-1}yz$$

$$= \begin{cases} [x, y]^t = y^{-1}x^{-1}yx = [y, x] \\ [xy, y^{-1}] = x^{-1}yxxy^{-1} \\ [x^{-1}, y] = xy^{-1}x^{-1}y. \\ [x, y][y, x] = x^{-1}y^{-1}xy \cdot y^{-1}x^{-1}yx = \text{id}. \end{cases}$$

Commutator of Typical Subgroups.
 D_{2n} , the dihedral of order $2n$.

S_n , the symmetric group on n letters

$$[S_3, S_3] = \langle [a, b] \mid a, b \in S_3 \rangle$$

$$[(1\ 2), (1\ 3)] = (1\ 2)(1\ 3)(1\ 2)(1\ 3) = (1\ 3\ 2)(1\ 3\ 2) = (1\ 2\ 3)$$

$$[(1\ 2), (2\ 3)] = (1\ 2)(2\ 3)(1\ 2)(2\ 3) = (1\ 2\ 3\ 1)(1\ 2\ 3\ 1) = (1\ 3\ 2)$$

$$[(1\ 3), (2\ 3)] = (1\ 3)(2\ 3)(1\ 3)(2\ 3) = (3\ 2\ 1)(3\ 2\ 1) = (1\ 2\ 3)$$

$$[(1\ 2), (1\ 2\ 3)] = (1\ 2)(1\ 2\ 3)^{-1}(1\ 2)(1\ 2\ 3) = (1\ 2)(3\ 2\ 1)(1\ 2)(1\ 2\ 3) = (1\ 3\ 2)$$

$$[(1\ 3), (1\ 2\ 3)] = (1\ 3)(1\ 2\ 3)^{-1}(1\ 3)(1\ 2\ 3) = (1\ 3)(3\ 2\ 1)(1\ 3)(1\ 2\ 3) = (1\ 3\ 2)$$

$$[(2\ 3), (1\ 2\ 3)] = (2\ 3)(1\ 2\ 3)^{-1}(2\ 3)(1\ 2\ 3) = (2\ 3)(3\ 2\ 1)(2\ 3)(1\ 2\ 3) = (1\ 3\ 2)$$

$$[(1\ 2\ 3), (1\ 3\ 2)] = \text{id}.$$

$$\text{Therefore, } [S_3, S_3] = \langle (1\ 2\ 3), (1\ 3\ 2) \rangle = \{\text{id}, (1\ 2\ 3), (1\ 3\ 2)\} = A_3.$$

Alternatively, since A_3 is normal subgroup of S_3 and $S_3/A_3 \cong C_2$, so $[S_3, S_3] \subseteq A_3$. Conversely, since $(1\ 3\ 2) \in A_3$ and $(1\ 3\ 2) = [(1\ 2), (2\ 3)] \in [S_3, S_3]$,
Therefore $[S_3, S_3] \supseteq A_3 = \langle (1\ 3\ 2) \rangle$.

ΓA_n , the alternating group.

Γ_8 . The Quaternion.